



**INNOVATION. AUTOMATION. ANALYTICS**

## **PROJECT ON**

# Hospital Patient Care Operations Analytics

**MySQL Project**

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## **Agenda:**

- Project Overview
- Objectives
- Database Design
- Tables and Relationships
- SQL Technologies and Concepts Used
- Advantages
- Challenges
- Conclusion

## Project Overview:

- This project is a **SQL-based Hospital Management and DataAnalysis System**.
- It uses **MySQL** to store and analyze hospital-related information.
- The database contains information about:
  - Patients
  - Doctors
  - Appointments
  - Treatments
  - Rooms
- SQL queries are used to analyze hospital operations and generate useful information.
- The project focuses on **patient activity, appointment demand, treatment performance, doctor workload, and room utilization**.

## Project Objectives:

The main objectives of the project are:

- To create a **structured hospital database**.
- To store patient and doctor information efficiently.
- To manage appointment details.
- To maintain treatment and room information.
- To analyze appointment demand across cities and services.
- To calculate treatment duration and waiting time.
- To analyze doctor and room performance.
- To identify treatment and appointment-related problems.
- To generate useful insights using SQL queries.

## Database Design:

- Database Name: **hospital**
- Database Type: **Relational Database**
- Database Technology: **MySQL**
- The database is divided into multiple tables based on different entities.
- Each table contains specific information.
- **Primary Keys** uniquely identify records.
- **Foreign Keys** establish relationships between tables.
- This structure makes data organized and easier to analyze.

## SQL Technologies and Concepts Used:

- **SELECT** – retrieve data
- **WHERE** – filter records
- **DISTINCT** – find unique values
- **COUNT()** – count records
- **SUM()** – calculate total values
- **AVG()** – calculate averages
- **GROUP BY** – group similar records
- **HAVING** – filter grouped results
- **ORDER BY** – sort results
- **JOIN** – combine tables
- **LEFT JOIN** – preserve records from the left table
- **Subqueries** – use one query inside another
- **CASE** – conditional analysis

## **Advantages:**

### **➤ Organized Data Management**

Hospital information is separated into structured tables.

### **➤ Easy Data Retrieval**

SQL allows required information to be retrieved quickly.

### **➤ Data Analysis**

We can calculate appointment counts, treatment duration, waiting time, and costs.

### **➤ Relationship-Based Analysis**

Connected tables allow us to combine patient, doctor, appointment, treatment, and room information.

### **➤ Performance Analysis**

Doctor workload and room usage can be analyzed.

### **➤ Problem Identification**

The project can identify multiple treatment attempts, cancellations, no-shows, and rescheduling patterns.

## Challenges Faced:

### ➤ Database Design

Designing appropriate tables and deciding which information belongs in each table.

### ➤ Establishing Relationships

Connecting tables correctly using primary keys and foreign keys.

### ➤ Writing JOIN Queries

Combining information from multiple tables required careful query writing.

### ➤ Data Aggregation

Using COUNT(), SUM(), AVG(), GROUP BY, and HAVING correctly.

### ➤ Complex Analysis

Some questions require information from several tables.

### ➤ Data Interpretation

After getting SQL results, the results need to be interpreted correctly to understand hospital operations.

## Conclusion:

The project uses **MySQL to manage and analyze hospital data**. It connects patients, doctors, appointments, treatments, and rooms to understand **hospital operations and performance**.